
What a Passive Power Meter Can Certify About AI Training

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Abstract

ABSTRACT PENDING.

1. Introduction

Compute governance proposals call for evaluations, audits, and the registration or prohibition of training runs above specified compute thresholds (Shavit, 2023; Sastry et al., 2024; Baker et al., 2025; Heim & Koessler, 2024); U.S. Executive Order 14110 briefly operationalised one such threshold, requiring the reporting of training runs above 10^{26} FLOP (The White House, 2023). Every such scheme presupposes that a regulator can tell when training occurs on monitored hardware. Detection from prover-owned telemetry is feasible but conditional on tamper-resistant read paths that do not yet exist (Rahman et al., 2026; O’Gara et al., 2025); active challenge-response protocols remove the trust anchor but require the prover’s participation (Monfared et al., 2026). A power meter outside the prover’s control is the one passive channel a verifier can read today against an uncooperative prover. This paper asks, as a theory-and-methods question: what can such a meter certify, with what methods, at what cost in verifier information, and against an adversary who knows the detector?

2. Related work

Four mature literatures own pieces of this problem. We take their tools off the shelf; what is new here is the question (compute governance), the adversary (a prover who knows the detector and reshapes the trace to evade it), and the channel (a single, filtered, low-rate meter outside the prover’s control). We locate the work against each in turn, and against the closest governance-specific detectors cited in Section 1.

Non-intrusive load monitoring. Inferring which devices or activities are running from a single aggregate power me-

ter is the founding problem of non-intrusive load monitoring (Hart, 1992), with three decades of disaggregation technique—edge/event detection, steady-state signatures, and state-space models (Zoha et al., 2012). Certifying training from one datacenter meter is this problem lifted from the household scale. The load-bearing difference is the adversary: classical NILM assumes cooperative or indifferent appliances and targets their steady-state or transient-edge signatures, whereas our load actively hides, and the signature we exploit is a quasi-periodic, frequency-wandering second-order structure rather than a switching event. The power-side-channel malware detector of Clark et al. (2013) is the same idea one level up in bandwidth, again without an evading adversary.

Spectrum sensing and low-probability-of-detection signals. Deciding whether a source is active, and what it is doing, from noisy observations—explicitly against a transmitter trying to stay below the detection floor—is the subject of spectrum sensing in cognitive radio and of low-probability-of-intercept/detection work in electronic warfare. Its canonical detector taxonomy, the energy detector versus the matched filter versus the cyclostationary feature detector, and the known trade-offs between them (the matched filter is optimal but needs a template; cyclostationary detection is robust to noise-power uncertainty where the energy detector fails at low signal-to-noise ratio) (Yücek & Arslan, 2009), is exactly the structure of our Rung-1 bake-off (Section 5 and Appendix A). This is the closest structural twin to the enterprise; the pieces it does not supply are the frequency-wandering line with an order-tracking front end (Dandawaté & Giannakis, 1994; Bonnardot et al., 2005) and an adversary whose evasion is schedule de-periodicisation at a measurable throughput cost (Section 6).

Power side-channel analysis. That computation is legible in power is the founding premise of hardware-security power analysis: differential power analysis (Kocher et al., 1999), its correlation refinement (Brier et al., 2004), and profiled template attacks (Chari et al., 2002) recover secret keys from a device’s consumption trace. We inherit the premise and invert the regime. Side-channel analysis assumes a proximate, high-bandwidth probe and a cooperative target running a known computation, and extracts a fine-grained secret; we assume a remote, low-bandwidth,

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Table 1. The claim ladder (draft from plan §1.3). Each rung answers a strictly more governance-relevant question at a stated cost in verifier information; each rung’s hard ceiling motivates the next. Within each rung we consider a benign (non-adaptive) and an adaptive (hiding) prover.

Rung	Claim	Additional requirement	Hard ceiling
1. Structural evidence	Order-coherent power variation exists under a specified structural null	External time-resolved trace only	Periodic physical confounds; structural-null misspecification; strong superposition or dilution; active power shaping; cadence moved outside the searched family; die-to-meter attenuation
2. Conditional classification	The trace is more training-like than the stated inference population	A representative or explicitly modelled inference null	Domain shift; the semantic decoy (training-shaped execution without retained training)
3. Active physical authentication	A load behind the meter responded to an unpredictable verifier challenge	Prover participation; a challenge-bound schedule	A challenge-aware dummy load; no semantic binding by power alone
4. Work verification	The physical response is tied to declared training state transitions	A sound, fresh transcript/proof and an explicit binding assumption	The remote-proof / co-location gap, unless the assumed primitive closes it

filtered meter and an adversarial target, and extract only the coarse structural fact of whether efficient iteration-structured training is present. The template-attack idea—build a generative model of the trace and test against it—is the honest precedent for benchmarking a corpus-free detector against a generator-based optimality ceiling, which we realise as a Whittle Neyman–Pearson ceiling for our Rung-1 detectors (Appendix A). A parallel corpus attacks the inverse problem of extraction, recovering model architecture or execution from GPU side channels (Maia et al., 2022; Naghibijouybari et al., 2018; Hu et al., 2020; Liu et al., 2016).

Covert communication and steganography. The de-periodicisation frontier (Section 6) is, formally, the warden-versus-covert-transmitter game: the prover trains covertly under a schedule-distortion budget while the verifier plays the warden. Covert communication theory—the square-root law of Bash et al. (2013) and its descendants—and the information-theoretic model of steganography (Cachin, 2004), which measures a stegosystem’s security by the relative entropy between cover and stego distributions, study exactly the trade-off between how constrained a hidden activity is and how detectable it remains. This is the literature that turns an empirical frontier into a statement about *every* verifier: bounding a distributional distance between the training-trace and null-trace distributions as a function of the adversary’s distortion caps any detector’s power, yielding a lower bound on the cost of hiding rather than an upper bound on what our particular detector misses. Its existing results do not cover our object—a physics-constrained training schedule observed through a filtered, low-rate meter—so the bound is not inherited but has to be derived. We derive it for one attack family against a fixed verifier in Section 3.3, which also explains why the familiar route through Pinsker’s inequality proves the converse of what a cost-of-

hiding claim needs. Its distortion half is a result; the step from distortion to cost is an assumption (Assumption 3.8), and tightening the threshold against the order-tracking class remains open.

Challenge-response authentication. Authenticating a physical device by its response to an unpredictable query is the premise of distance-bounding protocols (Brands & Chaum, 1994) and physical unclonable functions (Pappu et al., 2002), and, in compute governance specifically, of the active challenge-response telemetry that trades the passive channel for prover participation (Monfared et al., 2026). Rung 3 (Section 8) is this idea applied to the training-verification question over a filtered, low-rate power meter: the verifier correlates external power against a fresh challenge, and the null comes from re-randomising it rather than from a matched-hardware corpus. What we add to the pattern is the channel and the adversary—a remote meter and a prover free to answer with a challenge-aware dummy load—and, unlike device authentication, we do not build the cryptographic binding that would tie the response to declared work; we state it as an assumed ideal functionality (Section 8) and leave its construction open.

3. Threat model and identifiability

3.1. The observation channel

The verifier holds one scalar time series from a meter outside the prover’s control. The minimum model separates the device from the channel:

$$P_{\text{meter}}(t) = (h * P_{\text{device}})(t) + P_{\text{other}}(t) + \eta(t), \quad (1)$$

$$P_{\text{device}}(t) = r(t) F(t) + P_0(t), \quad (2)$$

where $F(t)$ is the instantaneous operation rate, $r(t)$ the energy cost per operation, $P_0(t)$ the non-compute overhead, h collects power-delivery dynamics, meter integration, reporting filters, and sampling, $P_{\text{other}}(t)$ is co-resident load behind the same meter, and $\eta(t)$ is measurement noise the prover cannot forge. The linear time-invariant form of h is an explicit first approximation; h may be operating-point dependent. Until simultaneous on-device and external-meter measurement is performed, claims about what reaches an off-chip sensor are predictions (Section 9).

3.2. The adversary: operators and budgets

The prover may apply, singly or in composition:

- **phase attack:** jitter, drift, or randomised work per iteration;
- **frequency relocation:** move the cadence toward or outside the search family;
- **additive burial:** independent workload superposition / spectral-floor jamming;
- **amplitude shaping:** added load flattening the compute/communication swing;
- **dilution:** reduce the training duty fraction among other work;
- **semantic decoy:** reproduce the physical schedule without retained training;
- **challenge-aware decoy:** a correlated dummy response in the active protocol (Section 8).

With unlimited superposition, dilution, shaping power, time, or freedom to restructure work, the prover can drive the signal-to-noise ratio arbitrarily low or synthesise the null; robustness claims are therefore made only over specified attack families at specified budgets.

3.3. Identifiability and the cost of hiding

Two questions have to be kept apart. *Detection* asks whether this trace is training rather than the null, and its tools are likelihood-ratio and cyclostationary tests. *Cost of hiding* asks how far the prover must distort its schedule before the training-trace *distribution* becomes indistinguishable from the null *distribution*, and its tools are distributional distances. The distinction matters because our empirical results bound the wrong direction for a governance claim: a detection sweep says how much can be hidden from *our* detector, which a better detector moves. This subsection states what we can prove in the other direction. Proofs are in Appendix B.

The first result is negative and sets the ceiling for the whole passive channel. Write w for a computational execution, h for the hardware and power-delivery path, and $K_h(w)$ for the law of the observed trace that Equations (1) and (2) induce.

Definition 3.1 (Observational equivalence). Two executions are observationally equivalent when $K_{h_0}(w_0) = K_{h_1}(w_1)$.

Proposition 3.2 (No passive semantic identification). *For (h_0, w_0) and (h_1, w_1) equivalent in the sense of Definition 3.1, every detector $\phi(P_{\text{obs}})$ declares training with the same probability under both. Consequently no passive scalar-power detector can certify the semantic content of a computation against an unrestricted adversary: a non-vacuous guarantee has to put the observationally equivalent pairs out of scope. Four restrictions do that — on the admissible non-training workload class, on the adversary’s freedom to add or restructure computation, on the hardware and power-transfer family, or on the physical property claimed — while a fifth route, changing the protocol, leaves the passive channel altogether (Section 8). We do not claim the list is exhaustive.*

The construction behind it is nearly available on the prover’s own hardware: a program can execute the same forward, backward, and optimiser kernels as training and discard each updated state, making its trace arbitrarily close to genuine training while its semantic classification differs. This is not a hypothetical weakness. It is what Section 7 measures: five non-training loads, the discarded-update decoy among them, score as training under all three decision rules. Proposition 3.2 says that outcome is a property of the channel rather than of our classifier, which is why Rung 2’s ceiling cannot be removed by a better detector and why Rung 3 has to add a challenge.

What remains provable is quantitative and concerns the *cost* of hiding. It is stated over the target population the paper scopes itself to, realised by the generator of Section 4:

Definition 3.3 (Admissible training family). Efficient, synchronous, iteration-structured training whose compute/communication alternation exposes a recoverable cadence f_0 . Within it, the synchronous exchange occupies a down-phase whose duration does not scale with the work accumulated in the preceding up-phase.

Definition 3.4 (ε -covertness). An attack is ε -covert against a class of verifiers if no test in the class attains detection advantage (true-positive minus false-positive rate) above ε . Since any single test’s advantage is at most the total variation $\text{TV}(P_{\text{train}}, P_{\text{null}})$ between the two laws (Tsybakov, 2009), covertness against *every* verifier requires that total variation to be at most ε .

Consider the concrete attack the frontier prices: i.i.d. fractional period jitter of scale σ , so iteration i has period

$1/(f_0(1 + \xi_i))$ with $\xi_i \sim \mathcal{N}(0, \sigma^2)$. Such jitter *accumulates*: the iteration-boundary phase performs a random walk.

Proposition 3.5 (Line attenuation under phase diffusion). *Under i.i.d. fractional period jitter of scale σ the boundary phase diffuses with coefficient $D \approx (2\pi)^2 \sigma^2 f_0$, and the coherent power retained in one frequency bin of a record of length T obeys*

$$\frac{P_c(\sigma)}{P_c(0)} \approx \frac{1}{1 + \kappa \sigma^2}, \quad \kappa = O(\pi f_0 T). \quad (3)$$

Proposition 3.6 (Necessary distortion for covertness). *Let $J(\sigma)$ be the advantage attained by the fixed-bin threshold test at the nominal cadence, which decreases in σ through Equation (3). Because $J \leq \text{TV}$, any σ with $J(\sigma) > \varepsilon$ is not ε -covert. Hence ε -covertiness requires $\sigma \geq \sigma^*(\varepsilon)$, where σ^* solves $J(\sigma^*) = \varepsilon$; it is strictly positive and increases as ε falls. Enlarging the verifier class can only raise $\sigma^*(\varepsilon)$, so the bound obtained from any one exhibited verifier is valid but not necessarily tight.*

Remark 3.7 (Which inequality is doing the work). Proposition 3.6 runs through an *explicit* verifier, and that is essential. The familiar covert-communication route — Pinsker’s $\text{TV} \leq \sqrt{\text{KL}/2}$ (Tsybakov, 2009; Cover & Thomas, 2006) — bounds total variation from above, so forcing that bound below ε would establish that some distortion *suffices* for covertness, not that any is *necessary*. Only a verifier that demonstrably attains advantage J certifies that smaller distortions fail. Note what this does and does not cost us. Because $J \leq \text{TV}$, a σ ruled out by the fixed-bin test cannot be ε -covert in the total-variation sense, and so is ruled out against any verifier class that contains that test — the five-detector bank of Section 6, order tracker and all. What it does not settle is a class that omits the fixed-bin test: TV bounds such a class’s advantage from above, not below. What it is also not is tight. A stronger verifier attains a larger advantage at each σ and so yields a larger σ^* , and since our own experiments show the tracking class to be the stronger one, the threshold we report is a loose reading of the true requirement rather than a limit on where it applies.

Distortion is not yet cost. The bridge is a physical property of the admissible family in Definition 3.3, and we state it as an assumption rather than a result, because two steps in its justification are not closed (Appendix B).

Assumption 3.8 (Cost of hiding). Within Definition 3.3 and at fixed total useful work, reaching $\sigma^*(\varepsilon)$ costs throughput or learning efficiency; it cannot be free in both. Period jitter then has only two realisations. Inserting idle time inflates the mean period and pays in throughput. Varying the real work per iteration avoids idling, but the down-phase duration does not scale with the accumulated work and that work is quantised, so a component at the exchange cadence survives; removing it requires either idling or desynchronising the

exchange, and the latter changes the optimisation being performed.

Remark 3.9 (Why this is an assumption). The surviving exchange-cadence component is recovered by the order-tracking class, not by the fixed-bin verifier of Proposition 3.6: under variable real work Section 6 measures the fixed tests below chance. So the argument establishes that a work-varying prover stays visible to a *tracker*, not that it must pay to become ε -covert against the verifier whose σ^* we computed. And fixing total useful work is a real restriction: substituting useful non-training work (the additive-burial and dilution families of Section 3.2) modulates the period while idling nothing and degrading no optimisation. The disjunction is therefore asserted over timing attacks at fixed work, and the measured support for it is the comparative one — at matched total distortion, work variation is the more detectable of the two at every $\sigma \geq 0.05$, the two being indistinguishable within sampling error below that (Figure 12).

Remark 3.10 (Scope of the bound). Three limits fix the wording. The cost in Assumption 3.8 is a disjunction, not a figure: its throughput leg is anchored to measurement (Section 6), while its learning-efficiency leg is argued and remains this paper’s central open empirical question. The advantage J of Proposition 3.6 is measured between the two laws of our own generator, so the bound is conditional on that generator standing in for the workloads it models. And the claim is that hiding *to a stated* ε must be paid for — expressly not that de-periodicisation is inefficient in general, which Section 6 refutes by exhibiting a near-free attack that defeats every fixed test.

4. Scenario models

5. Structural evidence (Rung 1)

Rung 1 asks the weakest question on the ladder: does the trace carry order-coherent power variation that a stated structural null does not produce? Rejecting the null establishes *coherent repeated power structure*, not training: reading that structure as repeated multi-phase computation additionally requires the nuisance model of Section 3 that excludes periodic physical confounds, of which the measured A100 power-controller oscillation (Section 9) is the standing counterexample. Rung 1 is evidence, not identification. We treat two prover settings in turn: a benign prover whose cadence is stationary, and an adaptive prover who lets the cadence wander to hide it. The benign case is easy, and the paper is not about the benign case; the adaptive case is the methodological heart, and the burden of this section is to show that its detector has a defensible null.

5.1. Benign setting: the qualified baseline

When the cadence is stationary, a repeated iteration structure appears as a line, or a harmonic comb, in the power spectrum, and the natural test is the Thomson multitaper harmonic F-test (Thomson, 1982; Percival & Walden, 1993). At a fixed frequency f the statistic $F(f)$ is distributed as $F(2, 2K - 2)$ under a locally smooth Gaussian spectral null with K tapers, so the test is *approximately pivotal under a stationary, locally smooth spectral null*. It is not an exact false-alarm guarantee across arbitrary external power traces: coloured baselines, controller interference, and co-resident load all violate the local-smoothness premise. Because the prover's cadence is not known a priori, the test must search a band of candidate frequencies and harmonics, which requires a search correction.

The analytic correction is itself unreliable. Sweeping the padded frequency grid and applying a Bonferroni correction over the Rayleigh-resolution cell count is *anti-conservative by a factor of roughly 3.6–6×*, uniformly across our null suite: the maximum of F over the padded grid behaves like about five times more effective independent tests than the Rayleigh count, because the F ratio fluctuates below the Rayleigh scale. The analytic multitaper tail is therefore usable only as a ranking statistic; its operative false-alarm calibration is the Monte-Carlo one supplied by the harness of Section 5.3. A spectral matched filter (peak PSD at the searched cadence) serves as the simple comparator. Both are honest baselines for a stationary line, and both fail the moment the line wanders: as the bake-off of Appendix A shows, the fixed multitaper F-test retains essentially no usable detection power once the cadence drifts, which is exactly why the adaptive setting needs a different detector.

5.2. Adaptive setting: tracked cyclostationary detection

An adaptive prover breaks the benign baselines cheaply by letting the cycle frequency wander, smearing the line across the searched band. We restore coherence by estimating the wander and undoing it before testing. The detector has three components, each with one job. First, a tracker estimates the instantaneous cycle frequency $\hat{f}_0(t)$ from the time–frequency plane; the tracker is a nuisance-parameter estimator, *not* the detector. Second, the phase estimate

$$\hat{\theta}(t) = 2\pi \int_0^t \hat{f}_0(s) ds \quad (4)$$

defines a resampling of the trace at equal increments of $\hat{\theta}$ (tachless order tracking (Bonnardot et al., 2005)), so a wandering line becomes a constant-order line. Third, the de-warped signal is tested for cyclostationarity at candidate cycle frequency α with the Dandawaté–Giannakis statistic (Dandawaté & Giannakis, 1994)

$$Q_\alpha = N \hat{r}_\alpha^H \hat{\Sigma}_\alpha^{-1} \hat{r}_\alpha, \quad (5)$$

where $\hat{r}_\alpha$ collects the estimated cyclic autocovariances at lags τ and $\hat{\Sigma}_\alpha$ is a consistent estimate of their long-run covariance. For a *fixed* α , Q_α is asymptotically χ_{2L}^2 under the structural null, given weak dependence, consistent covariance estimation, and regularity.

Those conditions do not automatically cover the complete adaptive pipeline. The tracker performs an effective search over frequency paths (a look-elsewhere effect); the time warp is estimated from the same record it tests; the covariance estimator $\hat{\Sigma}_\alpha$ must be chosen; and the asymptotic χ^2 tail may be a poor guide at finite length. Whether the null survives these steps — and in what form it can be claimed — is the make-or-break question of Section 5.3. We do not call the adaptive pipeline an analytic CFAR detector; Section 5.3 establishes what it is instead.

5.3. Null validity of the complete pipeline

The question is whether the adaptive pipeline has a null we can defend. We answer it with a staged design that isolates each adaptive ingredient in turn, crossed with a graded suite of nulls, and measured by the realised false-alarm rate at operationally small nominal levels — not only 0.05. The four stages add one adaptive step at a time: (1) a fixed cyclic frequency α with no resampling; (2) resampling under a *known* independent time warp; (3) a warp *estimated* on held-out blocks and scored only on the complement; and (4) the fully adaptive full-trace Viterbi tracker. The null suite grades from stationary Gaussian and coloured processes (white, AR(1), a resonant AR(2)), through stationary non-Gaussian processes (heavy-tailed, skewed, squared Gaussian), to locally stationary and controller-like processes (a Van der Pol-type power-management limit cycle mimicking the measured A100 controller). Each cell draws $M = 10^4$ null traces; realised rates carry Clopper–Pearson 95% intervals. Figure 1 shows the whole grid.

Level safety holds everywhere. Across the entire campaign no stationary Gaussian, coloured, or non-Gaussian null inflates the realised rate at any stage or level: the failure mode the gate was built to catch — adaptivity silently raising the false-alarm rate — does not occur. On the white null the fixed- α stage is near nominal (0.056, 0.012, and 0.0013 at nominal 0.05, 10^{-2} , and 10^{-3}); on coloured and non-Gaussian nulls the same stage is strongly conservative (realised rate ≈ 0). Tracing the three adaptive steps on the white null, where the fixed- α stage was near nominal, isolates what each one does:

- **Resampling alone (stage 1→2)** deflates the rate by about $3.4\times$ ($0.056 \rightarrow 0.016$ at nominal 0.05). Interpolating onto the angle grid correlates adjacent samples, the Bartlett estimator reads that as extra dependence and inflates $\hat{\Sigma}_\alpha$, and Q_α shrinks. Resampling *colours*

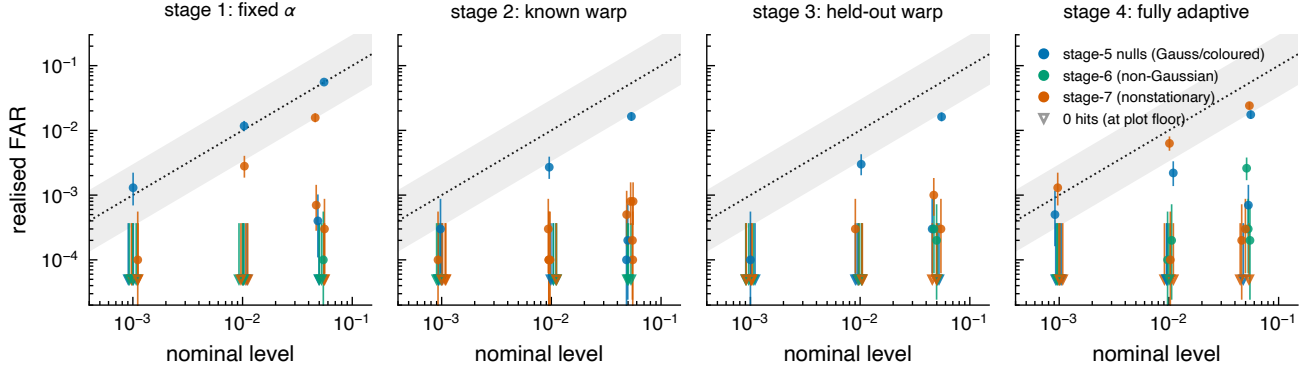


Figure 1. Realised versus nominal false-alarm rate for the adaptive pipeline (Dandawaté–Giannakis with a Bartlett HAC covariance), one panel per stage, every null in the suite, with Clopper–Pearson 95% intervals; nulls are coloured by stage group. No stationary null exceeds the nominal rate at any stage: the two-sided miss is *deflation*, never inflation.

the series.

- **Estimating the warp (stage 2→3)** moves the null nowhere ($0.016 \rightarrow 0.016$): the held-out design does what it promises.
- **Path selection (stage 3→4)** adds no significant inflation ($0.016 \rightarrow 0.018$; every interval pair overlaps). Even on the resonant AR(2) null, which the Viterbi tracker provably locks onto, the fully adaptive stage fires never in 10^4 traces: locking onto a broad noise resonance does not manufacture the phase-coherent cyclic structure Q_α requires.

The adaptive pipeline is therefore level-safe end to end. What fails is strict two-sided coverage: after resampling the pipeline is uniformly conservative, and the null p -values are not uniform (Kolmogorov–Smirnov uniformity is rejected at every adaptive cell).

No analytic calibration closes the deflation. We swept the open choices — record length, lag set, taper bandwidth, search band, interpolation scheme, and the covariance estimator — selecting on null-calibration quality across the suite, never on power. Only the covariance estimator moves the deflation, and it cannot be closed for the whole ladder at once (Figure 2). Widening the Bartlett bandwidth to $b = 54$ repairs the adaptive stage two-sided (AR(1) realised $0.050/0.012/0.0016$ across the three levels) — confirming that the deflation is an under-bandwidth artefact of the oscillating autocovariance of the resampled series — but the *same* bandwidth breaks the fixed- α anchor, inflating white to 0.073 at nominal 0.05 . No single covariance configuration in the swept family calibrates the ladder end to end, and Kolmogorov–Smirnov uniformity is rejected at every adaptive sweep point regardless. The asymptotic χ^2 tail survives only as a level-safe, power-costing conservative bound; it is not a calibration, and we do not report the adaptive pipeline as an analytic CFAR detector.

Surrogate calibration recovers the null. Running the entire pipeline — tracker, resampling, and test — inside a Fourier-phase surrogate null (which preserves the power spectrum and randomises phases, encoding the linear-stationary invariance) recovers exact calibration where the asymptotic tail was $3\text{--}6\times$ conservative. On the stationary Gaussian nulls the realised rate covers nominal across all three pipeline stages — fixed- α , held-out-warp, and fully adaptive (at nominal $0.05/10^{-2}$: white $0.051/0.009$, AR(1) $0.050/0.013$ fixed- α ; white $0.052/0.013$, AR(1) $0.054/0.010$ fully adaptive; the resonant-AR(2) and held-out-warp cells likewise) — and Kolmogorov–Smirnov uniformity is not rejected, the property the analytic p -values never achieved.¹ This is what “calibrated” means throughout the paper: an effective score with per-trace surrogate calibration, not an analytic guarantee.

The controller marks the attribution boundary. The surrogate null encodes linear stationarity, so a genuine limit cycle — which is itself cyclostationary — must and does inflate it: on the measured-controller null the surrogate-calibrated rate rises to $0.17\text{--}0.30$ at nominal 0.05 . This is not a defect. The detector correctly reports coherent cyclic structure in a controller trace; calling that structure *training*

¹Frozen on the MATS compute cluster over stages $1/3/4 \times$ eight nulls at the surrogate sizing $M = 2000$ null traces $\times$ $S = 999$ surrogates (the level-safety and bake-off numbers above are at $M = 10^4$; surrogate cells are capped lower because each carries S inner resamples). This supersedes the earlier partial $M = 1000/S = 199$ grid. The heavy-tailed but *linear* non-Gaussian nulls stay level-safe (conservative: t -innovation AR(1) $0.037/0.005$, log-normal $0.035/0.006$ fully adaptive); the one exception is the *nonlinear* squared-Gaussian null, which the fully adaptive stage over-rejects mildly ($0.087/0.018$, $\approx 1.7\times$, below the pre-registered $3\times$ threshold). Its cause is the same linear-stationarity assumption analysed next for the controller: a Fourier-phase surrogate cannot reproduce the quadratic phase structure of a squared process and so under-spreads the null. The level-safety claim is thus scoped to linear-stationary nulls.

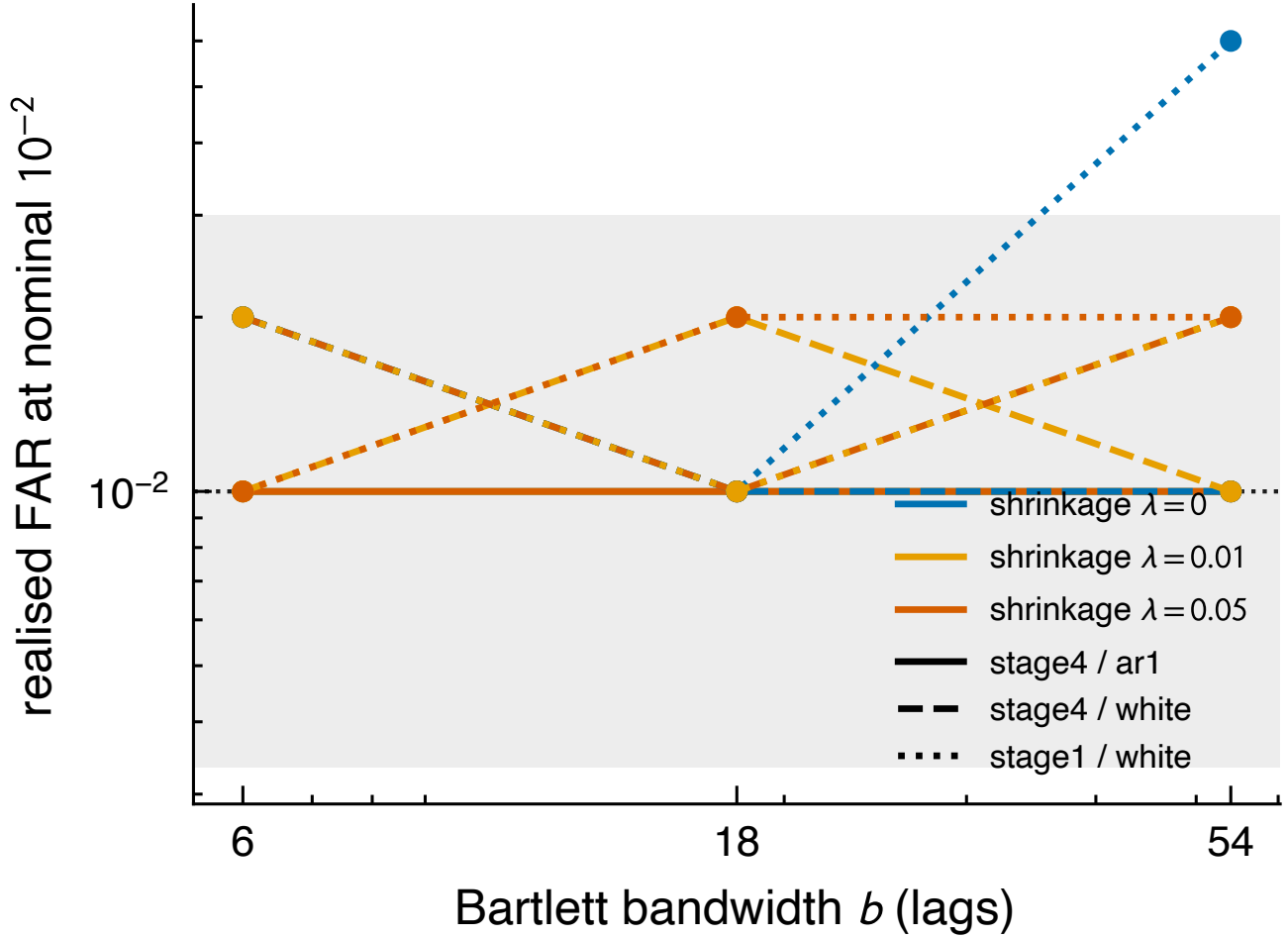


Figure 2. The covariance-estimator sweep is the fix-or-reframe axis: realised rate at nominal 10^{-2} versus Bartlett bandwidth b for each shrinkage. The bandwidth that repairs the fully adaptive stage (stage 4) inflates the fixed- α anchor (stage 1). No single configuration calibrates both.

is the causal-attribution step that needs the nuisance model of Section 3. The controller is the periodic physical confound that Rung 1 alone cannot exclude, and its false-alarm behaviour is exactly the ceiling the claim ladder assigns to this rung. The separation that Rung 1 *can* deliver — power against this confuser class, not merely against inference — is quantified by the bake-off of Appendix A.

6. The de-periodicisation frontier

Rungs 1 and 2 assume the prover leaves a recoverable cadence. An adaptive prover need not: it can de-periodicise its schedule to smear that cadence away. The question this section answers is not whether de-periodicisation is possible — it always is — but how far a prover must go before the meter loses the signal, and what that costs. This is a central result of the paper, not a caveat, and its shape is not the naive one. De-periodicising is not automatically cheap: some ways of doing it cost throughput without buying any

concealment, and the interesting question is whether *any* way is both cheap and effective.

We sweep six attack families as operators on the training schedule (Section 3.2) — per-iteration duration jitter, slowly drifting cadence, variable real work per iteration, cadence relocation toward the search-band edge, harmonic suppression, and dilution by co-resident load — plus a meter-degradation family treated separately in Section 6.1. For each family we step an attack level and, at every level, measure the physical de-periodicisation it induces (cadence coefficient of variation, phase diffusion) against detector power for the tracking class (the Viterbi tracker and the tracked Dandawaté–Giannakis order statistics) and for the fixed class (the spectral matched filter and the multitaper F-test), all at nominal false-alarm rate 0.05 with $n = 200$ traces per class. Where an attack has a measured systems cost we attach it from the single-A100 throughput anchors (Section 9); the plausible *learning-efficiency* penalty — batch-size, accumu-

lation, or update-timing changes that preserve tokens per second while worsening progress per token or per joule — is not something a synthetic power model can measure, and we carry it as the standing open empirical question rather than a result.²

The costly attacks are the wrong ones. The de-periodicisations that degrade the line most are exactly the ones a rational prover would not pay for. Idle-insertion jitter buys no concealment: it costs 15–375% throughput and leaves the tracked line detectable. Slowly drifting the cadence costs up to $\sim 140\%$ and is precisely the regime the order tracker was built to survive (Appendix A). Power shaping costs 170–230%. Against every one of these the tracking class holds while the prover pays a throughput tax that a compute-governance regime would itself flag.

The cheap attack collapses the fixed tests but not the tracker. The dangerous family is variable real work. Varying the useful computation performed per iteration moves the cadence without inserting idle time, so its throughput cost is essentially zero: at work-variation level 0.35 and 0.5 the measured overhead is -0.10% and -0.36% — zero within noise. At that price the fixed tests are gone. The spectral matched filter falls to detection rate 0.26 and 0.27 (its AUC collapses to 0.44 and 0.41, below chance), and the multitaper F-test sits at its floor. Yet the Viterbi tracker holds detection rate 1.0 at both levels (Figures 3 and 4). This is the genuine scope boundary of the passive channel: a prover willing to vary real work evades every cheap, fixed detector at no measurable systems cost, and only an order-tracking detector still sees the schedule.

The tracker’s own boundary is unmeasured. The Viterbi tracker does eventually bend — at work-variation level 0.7 its detection rate falls to 0.59 — but we do not report that as a defeat, because we cannot price it. The throughput cost of driving work variation that hard is not in the measured anchors, and the learning-efficiency cost is the open question above. The honest statement is therefore asymmetric: within the regime whose cost we can measure (overhead within noise) the tracker wins outright, and the far end of the frontier, where the tracker finally fails, is a cost we could not quantify rather than a cost shown to be small. That the tracker’s advantage here is real and not a tuning artefact is established independently in Appendix A: against the inference null the Viterbi tracker is essentially

Neyman–Pearson optimal, so the fixed tests’ collapse under variable work is a genuine loss of information from the schedule, not a deficiency of a particular detector.

The trade-off itself. Plotting hiding directly against measured cost (Figure 5) separates the two detector classes sharply. Against the fixed class the Pareto staircase reaches hiding 0.74 at essentially zero overhead and climbs only to 0.79 at 159% and 0.82 at 375% however much the prover spends: the cheapest attack is within 0.08 of the best one. Against the tracking class the staircase never leaves the floor, rising only to 0.16 and doing so at 159% overhead, so no attack in the anchored set buys meaningful concealment from the tracker at any price we can measure. Reading the two panels together is the paper’s central empirical claim in one figure: what the fixed tests lose for free, the tracker keeps at every measured cost. The figure also shows the boundary of that claim honestly. Only 16 of the 50 swept cells have a measured hardware cost anchor, and the rest — including work-variation level 0.7, where the tracker finally bends — carry no price at all, so they appear in a separate unpriced strip rather than being dropped.

Governance reading. A passive meter certifies efficient iteration-structured training against cheap, fixed tests only within a bounded de-periodicisation budget; a prover free to vary real work escapes those tests for nothing. Defeating the order tracker on top of that requires either a systems and learning cost we could not measure or a degradation of the meter channel itself (Section 6.1). The results of this section are therefore necessary conditions and an upper bound on hideability rather than a proof of training: they bound what a prover can hide from a fixed detector cheaply. The complementary direction — a lower bound on what hiding must cost, which no detection sweep can supply — is available for one attack family against a fixed verifier (Proposition 3.6) under the cost assumption of Assumption 3.8, and the distortion it requires can be priced against the same anchors used here. The tracking class dominates the fixed class at every cell we measured, so that threshold is a valid — but loose — reading of what a tracker would require: it is read off the weaker detector, so it understates the true requirement. Establishing it against the tracking class directly, rather than inheriting it from the measured dominance, remains open.

6.1. A minimum meter specification

The frontier above assumes a meter that resolves the cadence band. The verifier’s instrument is itself an attack surface: a prover who cannot de-periodicise cheaply may instead argue for, or be handed, a meter too coarse to see the line. We therefore sweep the meter rather than the schedule — sample rate and integration window between the nominal 20 Hz channel and a 1 Hz integrating sampler, plus

²Frozen on the MATS compute cluster (slurm array 5635), $n = 200$ per class over 50 cells, seed 0, the full five-detector set; `results/st2/frontier-summary.json` (provisional=false). The systems-cost axis reads only the clean throughput-overhead field of the measured campaign, not its signature-B line-band arrays, so the frontier is unaffected by the transport caveat of Section 9.

ST2 de-periodicisation frontier (% = measured throughput overhead)

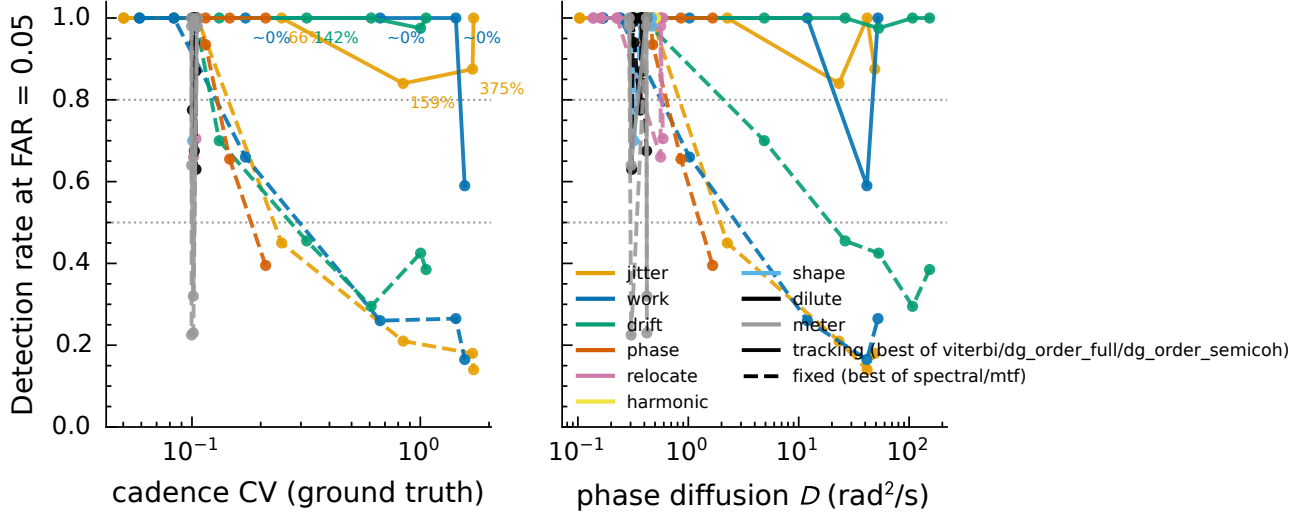


Figure 3. The de-periodicisation frontier: physical de-periodicisation (cadence coefficient of variation) against best tracking and best fixed detector power at nominal false-alarm rate 0.05, over all attack families. The near-vertical drop of the fixed class at the variable-real-work family, at essentially zero throughput cost, while the tracking class holds, is the scope boundary; costlier families (idle jitter, drift, shaping) sit at high throughput overhead without erasing the tracked line.

transfer-function notch depth and position near the cadence — with honest training against the hard inference null, and locate where each detector class dies.³ The result is a minimum specification the channel must meet for the Rung-1 and Rung-2 claims to hold, which we state as a channel requirement, not an attack cost.

The specification. The tracking class holds full power (detection rate $\gtrsim 0.9$ at nominal 0.05) if and only if the meter satisfies three conditions together (Figure 6). First, a sample rate of at least about 2 Hz: rates of 20, 10, 5, and 2 Hz all hold the Viterbi tracker near 1.0, 1 Hz collapses it to 0.36 as the Nyquist edge crosses the band, and 0.5 Hz is dead (every detector at chance). Second, an integration window no longer than about 0.5 s: at 20 Hz sampling a window of 0.5 s still holds (0.99), but a 1 s boxcar sinc-nulls the 1 Hz band centre and collapses the tracker to 0.25 even at full sample rate. Third, no deep transfer-function notch at the cadence: an off-band or shallow notch is tolerable (tracker ≈ 1.0), but a full-depth notch at 1.0–1.2 Hz roughly halves tracker power (to 0.48–0.55) without erasing it (Figure 7). The robustness ordering across the grid is Viterbi $\geq$ spectral $>$ the DG order family, which is the most sensitive detector

³Frozen on the MATS compute cluster (slurm array 5498), 48 cells, $n = 200, 300$ s traces; results/st2/meter_boundary_summary.json. Marginal and dead-cell AUCs are sensitive to the BLAS build, but the live-region boundary is not.

at the reference channel but the first to die as the channel degrades; the multitaper F-test is at its floor throughput.

The frontier’s meter-erasure point, decomposed. This sweep also resolves the single integrating-1 Hz point reported on the frontier, where even the tracker degrades (Viterbi detection rate 0.32, every other detector near chance). That point superposes two independent channel defects, and the boundary map shows either one suffices: at the reference channel the tracker is at 1.0; adding the 1 s integration alone drops it to 0.25; the 1 Hz sampling alone drops it to 0.36; the two together give 0.30. The meter-erasure regime is over-determined — it is a statement about the instrument, and a demand on it, rather than a de-periodicisation attack the prover can mount for free.

7. Conditional training classification (Rung 2)

Rung 1 asks whether coherent iteration structure is present. Rung 2 sharpens this into a decision: is a trace more training-like than a stated inference population? A decision needs a reference, and ours is the modelled continuous-batching inference null (Section 4); every Rung-2 conclusion is conditional on that null. We fix the task in advance — the feature vector, the decision rules, and the reporting — so that the classification result is not retrofitted to the answer.

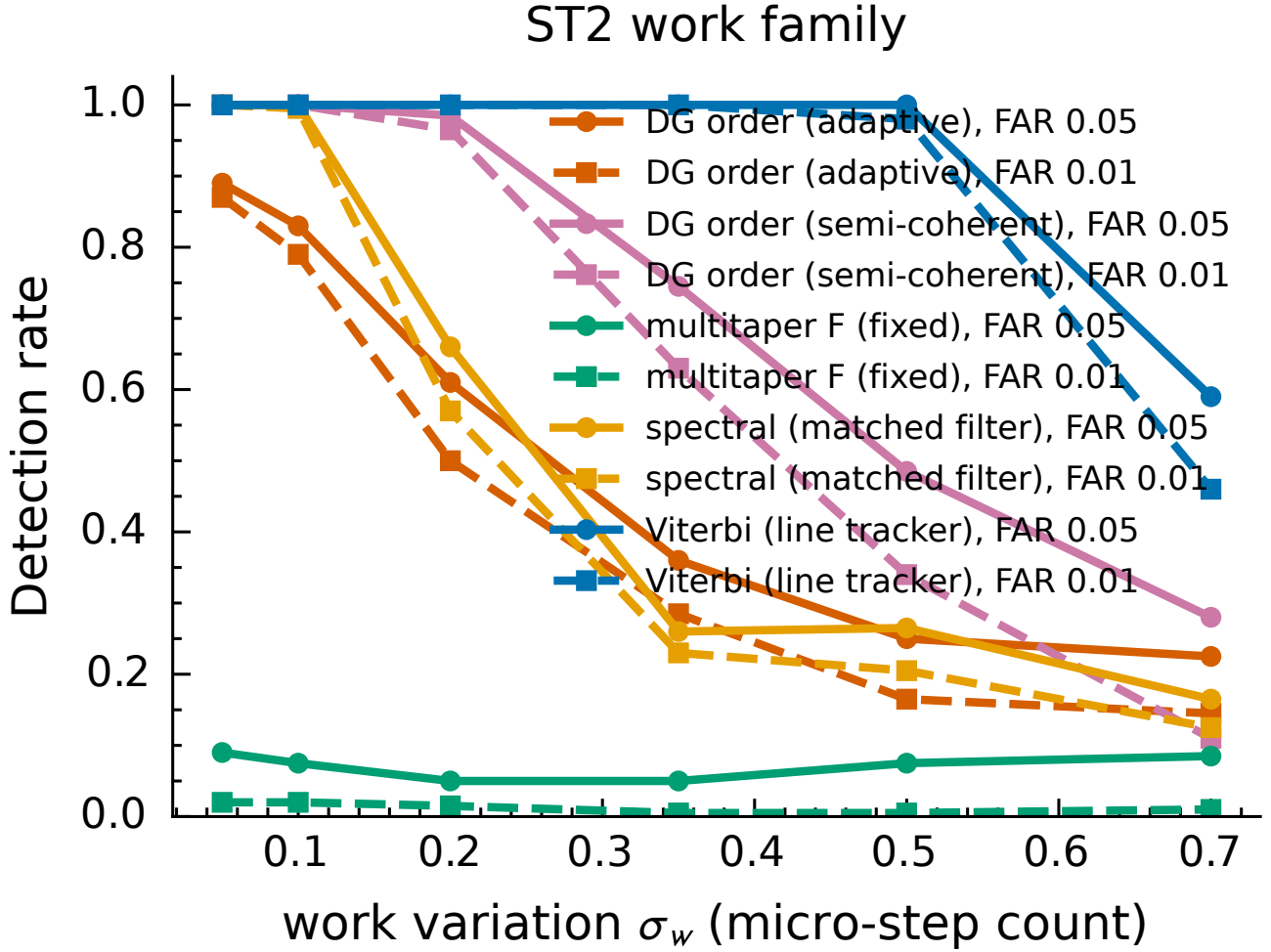


Figure 4. Detector power versus variable-real-work level for each detector. The fixed matched filter and multitaper F-test cross below chance by level 0.35; the tracked order statistics degrade more slowly; the Viterbi tracker holds full power until it finally bends at level 0.7 (detection rate 0.59) — a level whose throughput and learning cost we do not measure.

Features and decision rules, specified in advance. Each trace is reduced to an eight-dimensional physics feature vector estimated *from the trace alone*, with no oracle knowledge of the cadence: the six Rung-1 detector scores (multitaper F, harmonic comb, spectral matched filter, Viterbi, fixed- α DG, and tracked-order DG), plus two stability features read off the estimated Viterbi path — its frequency-path stability and its phase-folded waveform repeatability. On this fixed vector we compare three decision rules of increasing freedom: (i) a prespecified physics score, a fixed linear combination with the fixed-cadence comparators pre-registered out; (ii) a logistic discriminant fitted to the scenario generator and scored out-of-fold; and (iii) a random-forest learned reference, included to expose how much separation comes from the chosen null rather than from the structural method. We report area under the ROC curve and the false-negative rate at a false-positive rate of 0.05, kept separate from the

Rung-1 structural false alarms of Section 5.3.⁴

Stated population and transfer. On the stated population all three rules separate training from the hard inference null perfectly: area under the curve 1.00 with zero false negatives at a false-positive rate of 0.05 (Figure 8). The separation is not brittle to the population. Across six domain shifts — a low and a high cadence peak, a shortened 150 s record, a cadence band shifted to 0.8–1.8 Hz, a hostile controller-limit-cycle meter, and heavy coloured meter noise — the area under the curve stays 1.00. The physical schedule of efficient training is far enough from the inference null that leaving the stated parameter box does not close the gap.

⁴Frozen on the MATS compute cluster (slurm array 5612), $n = 200$ per class, 300 s traces, band 0.3–1.7 Hz, 14 cells (the stated population, six transfer shifts, and seven semantic controls); results/rung2/rung2-summary.json.

Cost of hiding: what measured adversary cost buys against each detector class

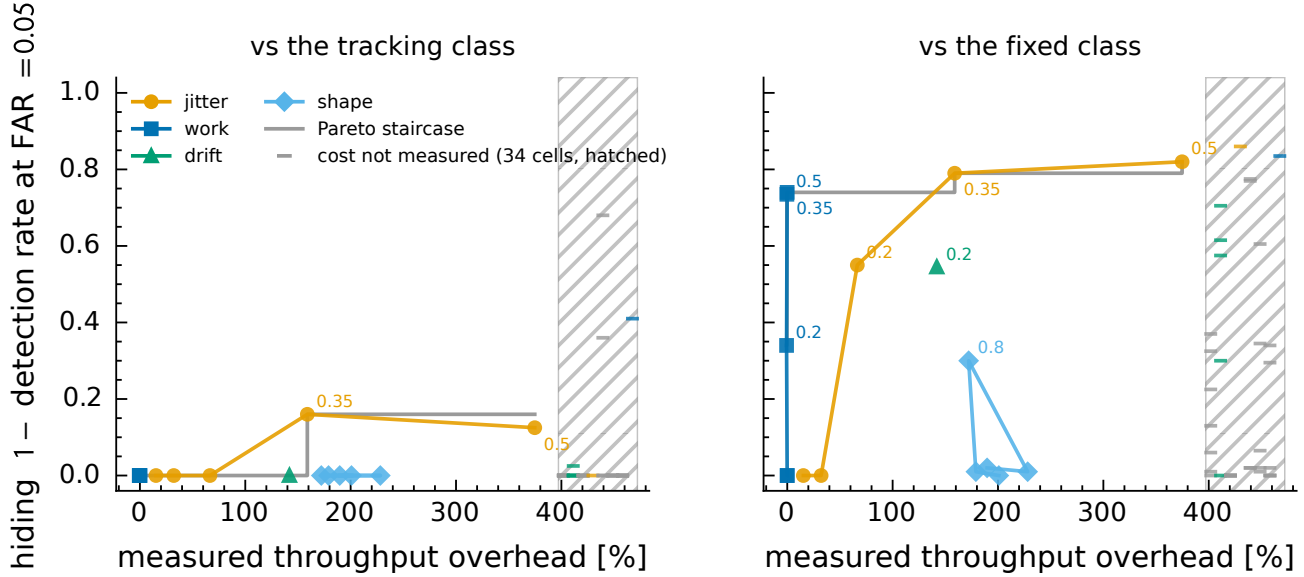


Figure 5. Cost of hiding: measured throughput overhead against hiding (1 – detection rate at false-alarm rate 0.05) for the four cost-anchored attack families, with the Pareto staircase — the best hiding available at or below a given cost. Against the tracking class (left) the staircase stays on the floor across the whole measured cost axis; against the fixed class (right) the variable-real-work family reaches hiding ~ 0.74 at zero cost. Cells whose attack level has no hardware cost anchor cannot be placed on the cost axis and are shown in the hatched strip, among them the level at which the tracker itself begins to fail. Attack levels are labelled where they hide something; the rest sit on the floor.

The semantic controls set the ceiling. The controls are the point of the rung. We built seven loads whose training status we know and asked what each rule scores (Table 2). Five non-training loads — a discarded-update decoy that runs the full forward/backward/optimizer step and throws the update away, a gradient-only computation, a training-shaped non-ML kernel loop, a controller limit cycle, and a periodic inference request generator — *all* score as training, at fraction 1.00 under every rule. A genuine training run diluted by an unrelated co-resident load is still caught (1.00). The reading is the one the claim ladder assigns to this rung: from power alone the verifier can certify a *training-shaped periodic compute schedule*, not that the computation is gradient-descent training rather than a mimic. This is the semantic decoy, and it is the ceiling that motivates the active protocol of Section 8. The controller cycle scoring as training is the same attribution boundary Rung 1 met in Section 5.3: the structural signal is real, and calling it training needs the nuisance model of Section 3.

Asynchronous training: a robustness result, not a miss. One control, strongly de-periodicised genuine training, was pre-registered as an expected miss — the case scoped out of Rung 2. At $n = 200$ it is not missed. The tracked physics rule catches it (area under the curve 0.89, 72% flagged)

and the learned reference catches all of it (100%); only the fitted linear discriminant is weak (0.75, 55%). This is consistent with the tracker-robustness story of Rungs 1 and 2: the asynchronous control de-periodicises by drifting the cadence centre and slipping the phase, which leaves a wandering but still coherent line — exactly the regime order tracking was built to survive. We therefore report it as a robustness result rather than the boundary. The genuine evasion is not slow wander but the $\approx$ zero-cost variable-real-work attack of Section 6, which moves the cadence *without* leaving a coherent line, and which is where the tracker actually bends.

8. From passive evidence to active authentication (Rungs 3–4)

Rung 2 certifies the physical schedule, not training semantics. Its ceiling is the semantic decoy: a load that reproduces the compute–communication cadence of training without retaining a trained model scores as training under every passive rule (Section 7). No passive test can separate the two, because the meter sees only the schedule and the decoy copies the schedule. The way past a passive ceiling is to stop watching and start asking. If the verifier perturbs the machine and the machine must react, a decoy that merely

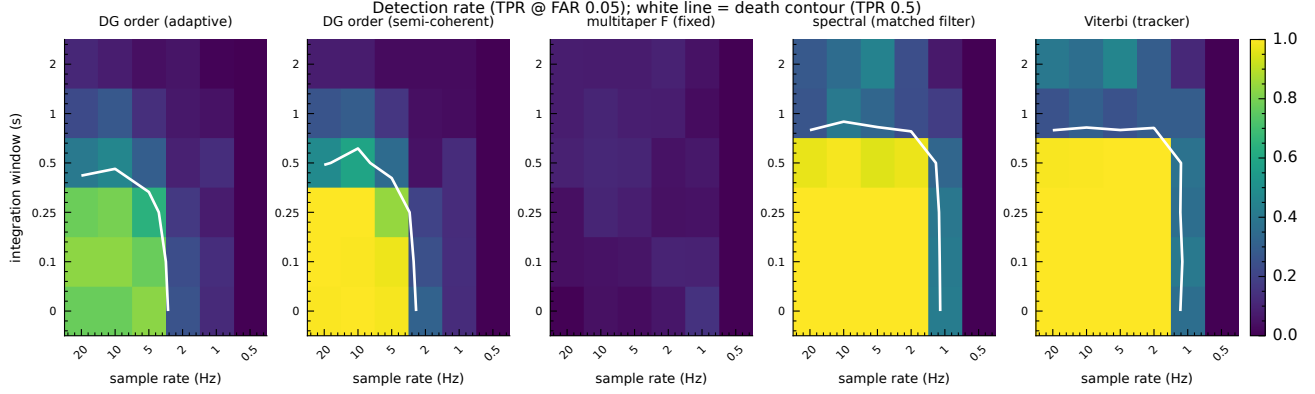


Figure 6. Minimum meter specification: detector power at nominal false-alarm rate 0.05 over sample rate $\times$ integration window, one panel per detector, honest training against the hard inference null. The live region is sample rate ≥ 2 Hz and integration window ≤ 0.5 s; below either edge the tracking class collapses toward chance.

Notch sub-sweep (20 Hz sampler): detection vs in-band anti-resonance

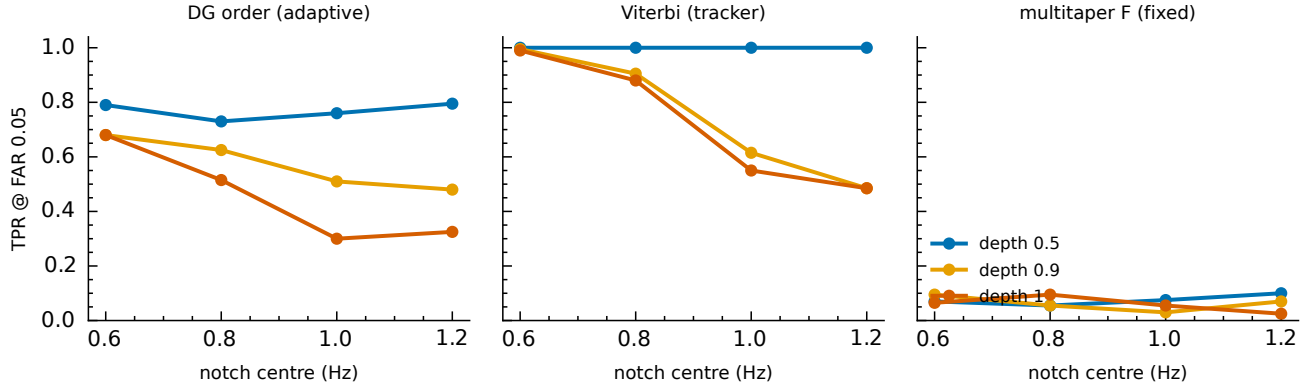


Figure 7. Transfer-function notch sub-sweep at 20 Hz sampling: tracker power versus notch centre frequency and depth. Only a deep notch placed at the cadence centre (1.0–1.2 Hz) materially degrades detection, and it halves rather than erases it.

replays a recording cannot follow: it does not know what it was asked.

Rung 3 makes this concrete with a challenge-synchronous protocol. The verifier emits an unpredictable balanced challenge sequence $c_k \in \{-1, +1\}$; the prover runs a committed, semantically-equivalent variation of its training schedule keyed to c_k — for instance a small pre-agreed shift in the boundary between two iterations that leaves the computation performed unchanged. The verifier correlates the external power trace against c_k over a causal lag window, in the manner of a lock-in amplifier or a spread-spectrum receiver recovering a known code from noise. The null needs no corpus and no matched hardware: it comes from re-randomising the challenge and recomputing the statistic. A positive result authenticates a physical fact — a load behind the meter responded to a signal it could not have predicted. This escapes the semantic decoy, whose passive schedule is by construction uncorrelated with the fresh challenge (Section 3.2). Its own ceiling is the challenge-aware dummy

load: a device that receives the challenge and mimics the prescribed response without training.

The correlation test is exact under the sharp null that the challenge has no effect on the measured process — but only conditional on the randomisation design, and the guarantee is fragile to the usual abuses (Fisher, 1935). Any maximisation the verifier performs, over causal lags, candidate impulse responses, or preprocessing choices, must be repeated inside every re-randomised replicate, or the reported level is optimistic. Carryover between challenge periods, any adaptivity in the prover’s schedule choices, and balance constraints on the challenge sequence all belong in the design that the re-randomisation samples from. We also flag the weaker reading a careless protocol invites: over a long enough record a correlation detector accrues the $\sqrt{N}$ processing gain of any matched receiver, so a positive result must be shown to certify the challenge binding and not merely that the verifier integrated for long enough.

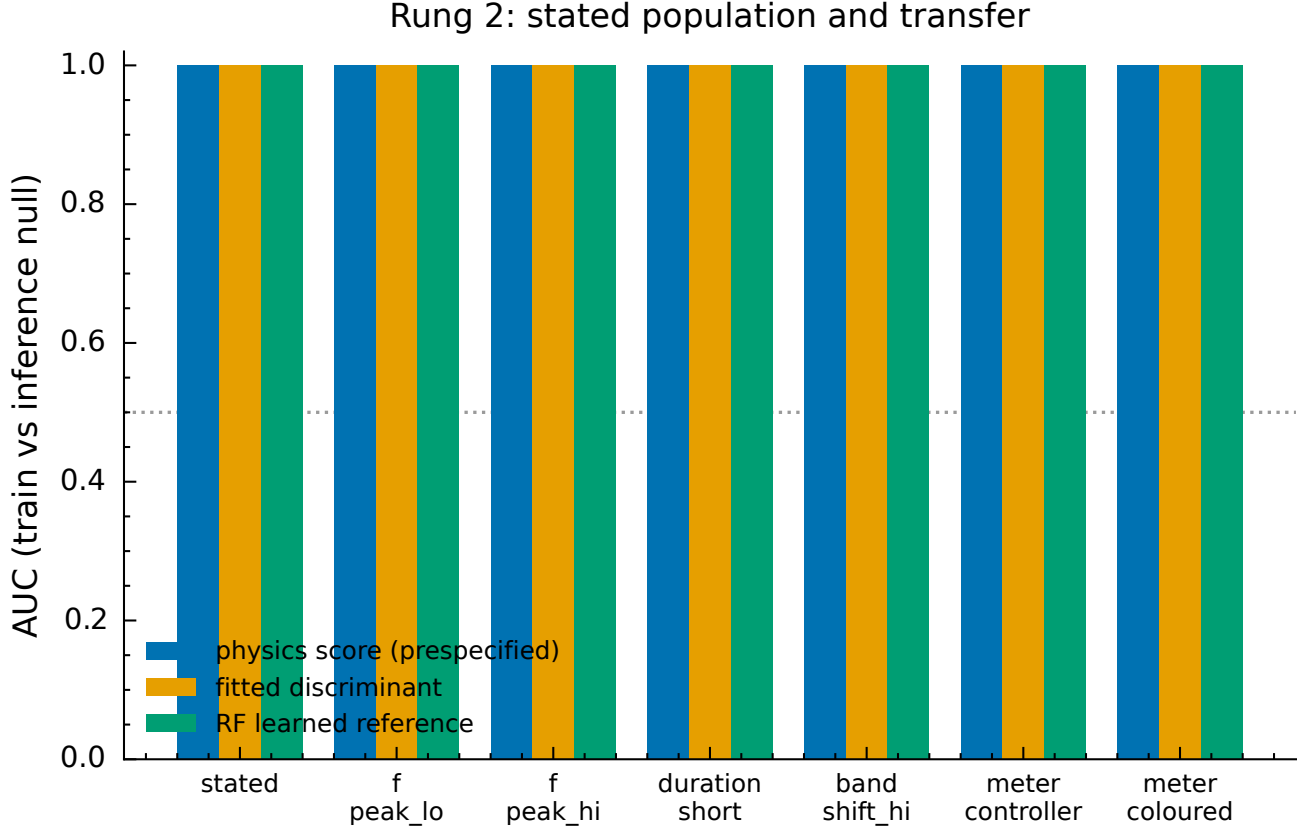


Figure 8. Rung-2 classification on the stated population (ROC / operating points for the three decision rules) and across the transfer shifts. All three rules reach area under the curve 1.00 on the stated population and hold it under every domain shift.

Rung 3 authenticates a physical response; it does not tie that response to training. Rung 4 adds that binding, and it is the rung this paper does not build. Rather than gesture at “a proof of training,” we state the missing piece as an ideal functionality (Canetti, 2001): a sound, fresh, challenge-bound transcript or proof that ties each challenge bit to the prescribed semantically-equivalent variation of declared training state transitions. Such a primitive must provide commitment before the challenge is revealed, unpredictability at commitment time, sound binding of the response to valid declared state transitions, freshness against replay, and an explicit position on whether the responding load and the declared computation are co-located. Proof-of-learning constructions (Jia et al., 2021) are the closest existing candidate, but they assume access to the training trace rather than an external meter. Conditional on such a primitive, challenge-synchronous power supplies physical evidence for the declared response; without it, the semantic work binding remains an assumed, open layer.

Two ceilings therefore remain. Rung 3’s is the challenge-aware dummy load; Rung 4’s is co-location: a genuine remote proof combined with a local challenge-correlated dummy still satisfies both the correlation test and the tran-

script unless the assumed primitive explicitly rules the separation out. Closing them empirically is future work, and it needs no cluster — challenge modulation is legible on a single card. A full Rung-3 study would fix a concrete pair of semantically-equivalent schedules, run the complete search inside every re-randomised replicate, pit the detector against a challenge-aware dummy-load adversary, and trace detection against modulation depth, throughput overhead, and observation time. We leave it as the active-verification direction the passive results here motivate but cannot reach — the response to the ceiling that Proposition 3.2 makes unavoidable for any passive detector (Section 10).

9. Reality check and transfer limits

Every result above is synthetic. The one place we hold the scenario model against measured hardware, it fails in an instructive way, and we report that failure as a result rather than a closing nod.

The negative transport case. On measured single-A100 traces the iteration cadence is present but shallow: a saturated card has no communication phase, so the iteration structure modulates a ~ 370 W draw by only a few watts

Table 2. Semantic-control outcomes: fraction of each load scored as training at a false-positive rate of 0.05, per decision rule. Five non-training loads all score as training; the only control the tracked rules degrade on is asynchronous (de-periodicised) genuine training, which they nonetheless mostly catch.

Control load	Training?	Physics	Fitted	Learned
Discarded-update decoy	no	1.00	1.00	1.00
Gradient-only	no	1.00	1.00	1.00
Non-ML kernel loop	no	1.00	1.00	1.00
Controller cycle	no	1.00	1.00	1.00
Periodic inference	no	1.00	1.00	1.00
Co-resident mixture	yes	1.00	1.00	1.00
Asynchronous training	yes	0.72	0.55	1.00

($\sim 0.4\%$ of board power). Worse for the naive reading, the spectra contain a 0.31–0.45 Hz component, roughly ten times stronger than the training line, that appears whenever the load varies in time but not at idle or under constant load — the signature of the card’s own power-management loop. It is a hardware line that says “time-varying load,” not “training,” and it is the standing counterexample that Rung 1 (Section 5) and the Rung-2 controller control (Section 7) are built to exclude.

Why this is the honest outcome. Two things make the single-card signature shallow. One saturated card has no synchronous communication barrier, so the deep compute–communication contrast the scenario model assumes is absent; and the generator is literature-parameterised, not calibrated to the measured channel, a gap of about $75\times$ in modulation depth. The synthetic results are therefore a sensitivity analysis over a plausible parameter family, not a prediction calibrated to this hardware — which is why the paper’s claims are scoped to the stated generator and its stated nulls throughout.

Falsifiable predictions. The model makes two predictions we cannot test here but state as falsifiable. Distributed synchronisation should deepen the line: at cluster scale the periodic all-reduce barrier produces a deep, coherent, legible cadence rather than the shallow single-card ripple, which is precisely the physics Assumption 3.8 leans on. And an external wall or rack meter should see a different, lower-passed trace than on-board NVML telemetry, shifting where the minimum meter specification of Section 6.1 binds. Both are predictions, not results.

10. Discussion and conclusion

We set out to ask what a passive power meter can certify about AI training, and organised the answer as a claim ladder in which each rung answers a more governance-relevant question at a higher cost in verifier information, and each rung’s ceiling motivates the next. Table 3 closes the ladder out with the paper’s evidence attached to each rung.

Scope. The results are fixed-horizon: we make no claim that the χ^2 statistic supports anytime-valid monitoring, and defer sequential e-process treatment. We use only the temporal-structure channel, excluding the amount-of-computation and energy channels. The cryptographic binding of Rung 4 is assumed as an ideal functionality, not built. Real distributed hardware enters only as the falsifiable predictions and the negative transport case of Section 9.

Limitations. The scenario model is literature-parameterised, not calibrated to measured hardware, so every quantitative result is a sensitivity analysis over a stated parameter family; the $\sim 75\times$ modulation gap of Section 9 is the standing reminder. The observation map is a linear time-invariant first approximation. The optimality ceiling of Appendix A is optimal only under the Whittle model, hence a lower bound on the true optimum. Every robustness claim is bounded by the attack families and budgets we specified; a family we did not model is outside it. The cost bound of Section 3.3 shares that limit: it covers i.i.d. period jitter, and while it holds against any verifier class containing the test it exhibits, it is obtained from a fixed-bin test and so reads loose against the order tracker our own experiments show to be stronger — it does not yet close the frontier’s far end. Its cost leg is an assumption (Assumption 3.8), not a result, for the reasons in Remark 3.9. And the learning-efficiency leg of that bound — the axis that would turn it into a single priced statement rather than a disjunction — is argued, not measured, and remains the central open empirical question.

Governance reading. What a regulator can buy today is graded. Rung 1 buys physical evidence of an efficient iteration-structured compute schedule from an external meter alone, robust to a wander-and-drift adversary but not to a prover who varies real work or degrades the meter. Rung 2 buys the further claim that the schedule is more training-like than a stated inference baseline — but not that the computation is training rather than a schedule-matched decoy, which is a limit of the passive channel, not of the method. Separating training from a decoy, and binding a physical response to declared work, both require moving off

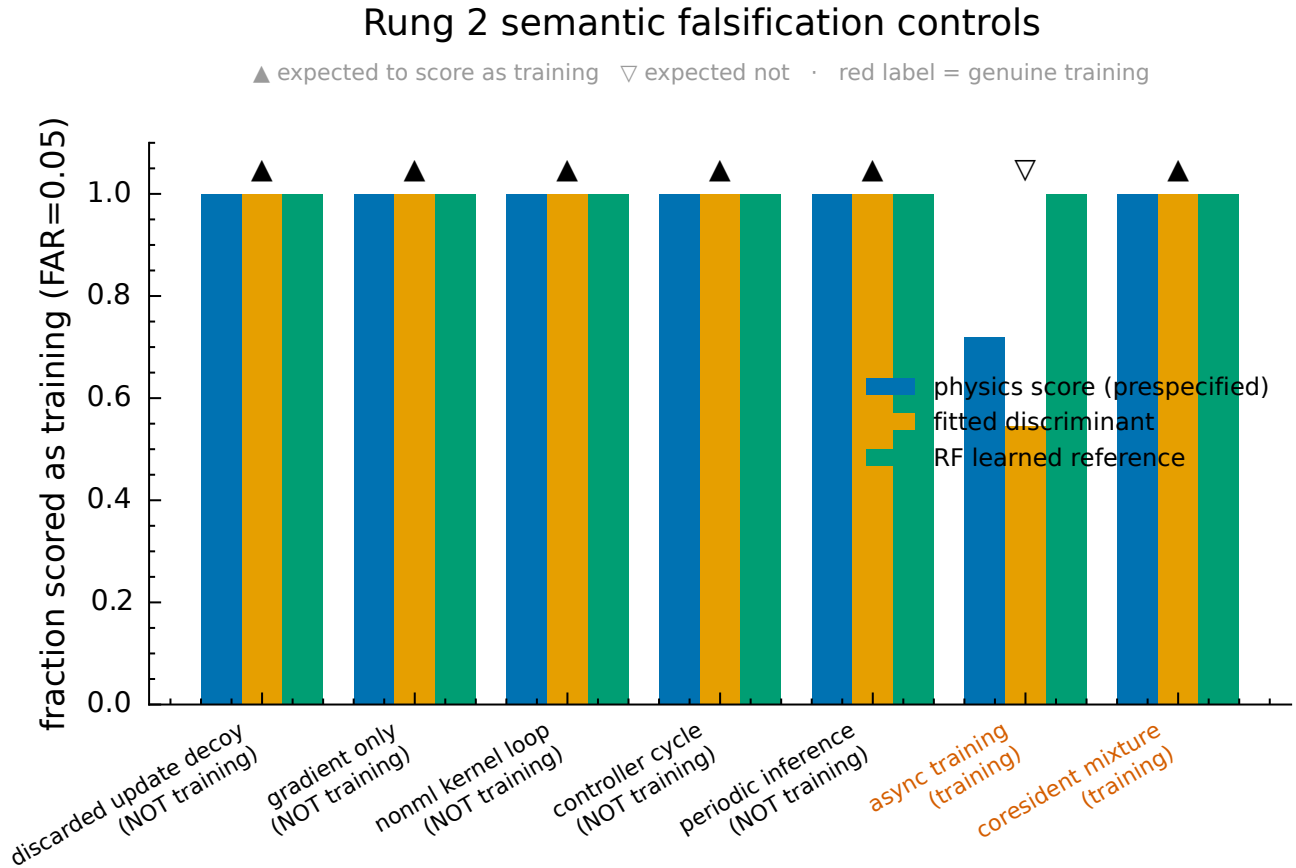


Figure 9. Semantic controls: fraction scored as training per decision rule. The non-training decoys are indistinguishable from training under every rule (the semantic-decoy ceiling); only asynchronous genuine training separates the rules, and the tracked physics rule still catches most of it.

the passive channel to the active protocol of Section 8 and the cryptographic primitive it assumes. The passive meter is thus a cheap, deployable front line that narrows the question and prices the cheap evasions, not a stand-alone proof of training.

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Table 3. The claim ladder closed out: what this paper establishes at each rung, where, and the hard ceiling it hits. Mirrors Table 1.

Rung	What the paper establishes	Evidence	Demonstrated ceiling
1. Structural evidence	Order-tracked cyclic structure is detected against stationary and controller-like nulls; the tracker is Neyman–Pearson optimal against the inference null	Section 5, Appendix A	Fast cadence wander (>0.8 Hz on 300 s); attributing structure to training needs the nuisance model; the meter must meet Section 6.1
2. Conditional classification	The trace is more training-like than the stated inference null (area under the curve 1.00, stable across domain shift)	Section 7	The semantic decoy: non-training loads with a training-shaped schedule score as training; for the discarded-update decoy Proposition 3.2 shows no passive detector can do better
Frontier	Fixed tests collapse under $\approx$ zero-cost variable real work while the tracker holds; costly de-periodicisations buy no concealment	Section 6	The tracker’s far end, at a systems and learning cost we could not measure
Cost of hiding	Hiding to detectability ε requires a distortion $\sigma^*(\varepsilon)$ that the measured anchors price at tens of percent of throughput, rising above 100% (i.i.d. period jitter)	Section 3.3, Appendix B	Read off a fixed-bin verifier, so valid against any class containing it, but loose; one attack family; internal to the stated generator; the throughput-or-learning disjunction is Assumption 3.8, not a result
3–4. Active authentication	A challenge-synchronous protocol and an assumed binding primitive, specified but not run	Section 8	Challenge-aware dummy load; the co-location gap

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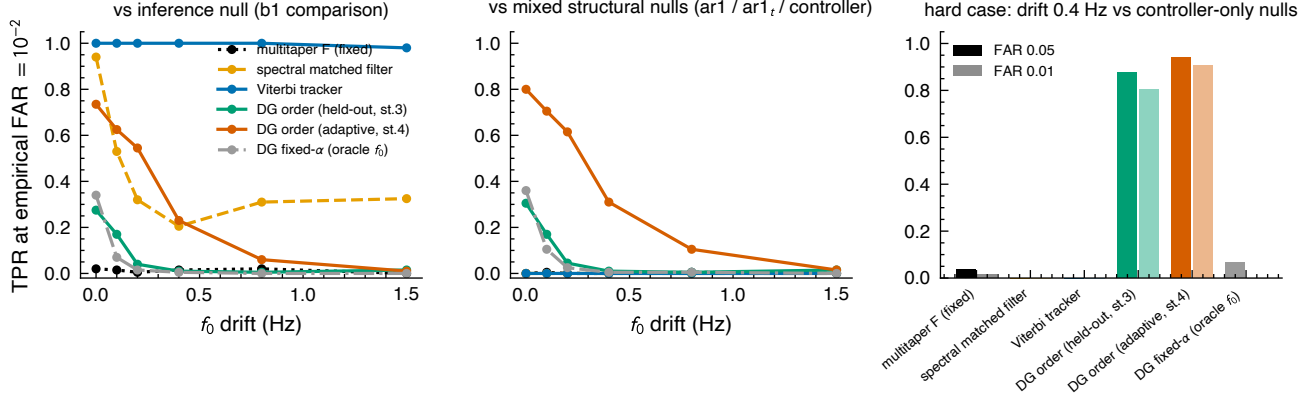


Figure 10. Detector bake-off on the cadence-drift axis: detection rate at empirical false-alarm rate versus drift, for each detector and both negative classes. Only the tracked order family retains power against the structural negatives; the spectral-shape detectors detect the controller third of the mix more strongly than the signal.

Table 4. Bake-off detection rate at nominal false-alarm rate 10^{-2} , by detector and cadence drift (Hz), against the inference null and the mixed structural population ($n = 200$ per class).

Detector	0	0.1	0.2	0.4	0.8	1.5
<i>vs. inference null</i>						
mt f (fixed multitaper F)	0.02	0.01	0.01	0.01	0.02	0.00
spectral matched filter	0.94	0.53	0.32	0.20	0.31	0.33
Viterbi tracker	1.00	1.00	1.00	1.00	1.00	0.98
DG order (split)	0.28	0.17	0.04	0.01	0.01	0.01
DG order (full)	0.73	0.62	0.55	0.23	0.06	0.01
DG order (semi-coherent)	1.00	1.00	0.97	0.56	0.07	0.01
DG fixed- α (oracle f_0)	0.34	0.07	0.01	0.01	0.00	0.00
<i>vs. mixed structural nulls</i>						
mt f	0.00	0.01	0.00	0.00	0.01	0.00
spectral matched filter	0.00	0.00	0.00	0.00	0.00	0.00
Viterbi tracker	0.00	0.00	0.00	0.00	0.00	0.00
DG order (split)	0.30	0.17	0.04	0.01	0.01	0.01
DG order (full)	0.80	0.70	0.61	0.31	0.10	0.01
DG order (semi-coherent)	1.00	1.00	0.99	0.69	0.14	0.03
DG fixed- α (oracle f_0)	0.36	0.10	0.03	0.01	0.01	0.00

A. Detector bake-off

The choice of detector for Rung 1 is an *outcome* of a pre-registered comparison, not an assumption. Before scoring any signal we fixed the detector set, the conditions, and the metric, so that “we present one method” is a finding rather than a choice. The detector set spans the natural alternatives: the fixed multitaper harmonic F-test (mt f); a spectral matched filter; the raw Viterbi tracker score; the fixed- α Dandawaté–Giannakis statistic given the *true* f_0 as an oracle; and three variants of the tracked order pipeline — split (held-out warp), full (fully adaptive pooled), and semi-coherent (a per-block Fisher combination). We score positives (Ko training traces) over a cadence-drift axis $f_0^{\text{drift}} \in \{0, 0.1, 0.2, 0.4, 0.8, 1.5\}$ Hz against two negative classes, each with its own empirical false-alarm threshold: the modelled inference null, and a mixed structural population (equal parts AR(1), heavy-tailed AR(1), and the measured controller). With $n = 200$ traces per class the 10^{-2} threshold rests on the second-highest negative score — a granularity we state rather than hide. Figure 10 plots the power curves; Table 4 gives the detection rates at nominal false-alarm rate 10^{-2} .

Three findings follow, and they are the reason the main text presents a single detector. First, *tracking is necessary*: the oracle fixed- α statistic, handed the true f_0 , collapses from 0.34 at zero drift to ≤ 0.1 under any wander — even the true frequency is useless to a coherent fixed- α test once the line moves — and the fixed multitaper F-test has essentially no usable power against either negative class at empirical thresholds. Second, *only the tracked order family separates the wandering line from structural confounds*. Against the mixed structural negatives the matched filter and the raw Viterbi score have detection

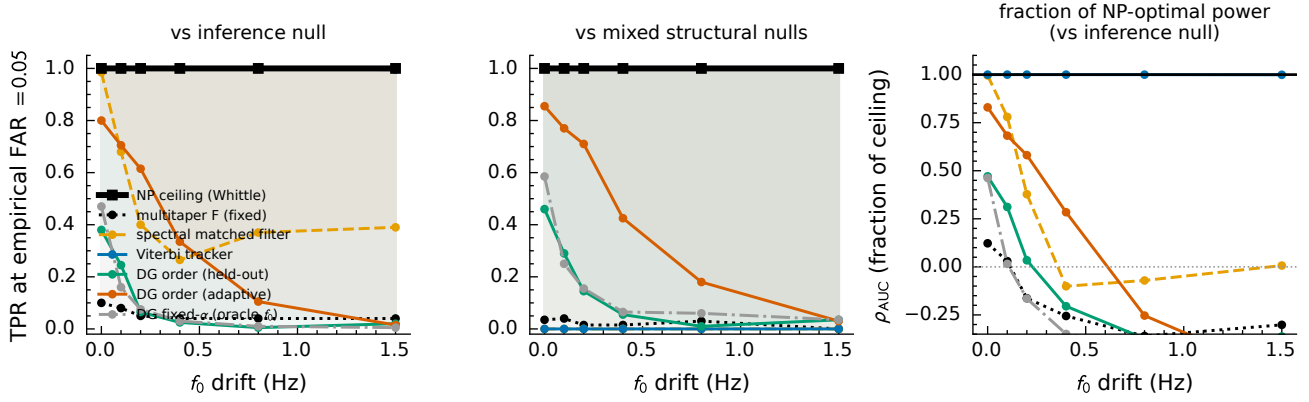


Figure 11. Corpus-free detectors as a fraction of the Whittle Neyman–Pearson ceiling versus cadence drift, against the inference null and the structural confusers. The ceiling is 1.0 everywhere; the Viterbi tracker saturates it against the inference null, while only the tracked order family approaches it against the structural confusers. The gap that opens at high drift is the headroom above current detectors.

rate zero at every drift: the controller third of the mixture fires them harder than the signal does, because a Viterbi score happily locks onto a limit cycle. The hard case makes this sharp — at drift 0.4 Hz against controller-only negatives the order pipeline holds 0.8–1.0 detection at 10^{-2} while every spectral-shape detector, including the fixed- α oracle, drops to near zero. Phase-coherent cyclic structure at the tracked order is the one signature the confuser cannot fake; this is the attribution boundary of Rung 1 drawn with power, not false-alarm rate. Third, the *semi-coherent variant dominates* the pooled statistic at every drift, filling the fast-wander power hole up to about 0.4 Hz while remaining level-safe.

The honest boundary is fast wander. Beyond drift 0.8 Hz every calibratable detector fades (the semi-coherent variant falls to 0.07–0.14 at 10^{-2}); only the raw Viterbi score keeps power against the inference null there, and it cannot tell a training run from a controller. A 300 s record does not resolve fast cadence wander with an honest null. This is a scope statement for the paper, not a repairable defect.

How far below optimal? The bake-off ranks detectors against each other; it does not say how good the winner is in absolute terms. Because we own the generators, we can benchmark against an optimality ceiling: an approximation to the Neyman–Pearson optimal test between the training and null generators. No closed-form likelihood exists — both are black-box samplers — so we use a Whittle spectral likelihood ratio on the in-band periodogram, with a per- f_0 Monte-Carlo training template bank marginalised over cadence, fitted on a disjoint stream and applied to the *exact* bake-off evaluation populations (parity check: maximum detection-rate difference 0).⁵ The ceiling is perfect: area under the curve and detection rate are 1.0 in every column at every drift (Figure 11). The training comb and the line-free nulls are separable in principle from a 300 s trace, so all the difficulty the bake-off exposes lives in the detector, not the problem.

Read as a fraction of this ceiling, two findings sharpen the bake-off. Against the inference null the Viterbi tracker is essentially Neyman–Pearson optimal: it holds the ceiling’s power across the whole wander axis, while the fixed matched filter’s fraction of attainable power collapses from 1.0 at zero drift through 0.78 and 0.38 to below zero by drift 0.4 Hz. Against the structural confusers and the hard controller-only case the tracker is *not* enough — only the tracked order family approaches the ceiling, with `dg_order_full` reaching 0.93 of it at low drift and fading to 0.10 by the fastest wander, while the Viterbi tracker and matched filter score zero (they lock onto the limit cycle). The widening gap to the ceiling at high drift is the honest headroom above today’s detectors, not a limit of the channel. The one caveat runs in our favour: the ceiling is optimal only under the Whittle model, a second-order stationary-Gaussian statistic that discards harmonic-phase coherence and non-Gaussian structure, so the true optimum can only exceed it. The perfect ceiling is therefore a lower bound on what is achievable, and reporting deployable detectors as a fraction of it is conservative in the right direction.

B. Identifiability details

This appendix proves the results stated in Section 3.3 and reports the numerical check that fixes their constants.

⁵Frozen on the MATS compute cluster (slurm job 5646), $n = 200$ per class, 61 cadence templates, 2000 Monte-Carlo draws each; results/st1/np.ceiling.summary.json.

Proof of Proposition 3.2. A passive verifier observes only P_{obs} , so its decision is a measurable function ϕ of that trace and its rejection probability is $\mathbb{E}_{K_h(w)}[\phi]$. If $K_{h_0}(w_0) = K_{h_1}(w_1)$ the two expectations coincide for every ϕ , so no test separates the pair, and no post-processing of the trace can (data processing cannot create distinguishability that the laws do not contain). The claim about restrictions follows by contraposition: a non-vacuous guarantee must exclude the equivalent pairs, and each listed lever excludes some of them. The amplitude degeneracy of Equation (2) shows the pairs are not exotic. The meter constrains only the product $r(t)F(t)$, so a workload with a different operation rate and a compensating energy cost per operation is already observationally equivalent, and temporal structure — not amplitude — is what the trace identifies. The near-constructive same-hardware case is the discarded-update decoy: forward, backward, and optimiser kernels executed on the declared schedule with each updated state discarded reproduces the compute/communication alternation, hence the trace, without retaining a model. The converse direction is equally open to the prover: genuine training can be run asynchronously, diluted among other loads, or actively power-shaped, so the equivalence classes are populated from both sides.

Proof of Proposition 3.5. Boundary i nominally carries phase $2\pi i$; fractional period jitter perturbs each boundary relative to its predecessor, so the *residual* of the boundary phase against a uniform clock accumulates independent increments and performs a random walk with $\text{Var}[\theta(t)] \approx Dt$ and $D \approx (2\pi)^2 \sigma^2 f_0$. For a Brownian phase the periodic component’s autocorrelation decays as $\exp(-D|\tau|/2)$, so its spectrum is a Lorentzian rather than a line, with half-width at half-maximum $\gamma = D/4\pi = \pi\sigma^2 f_0$ in Hz. A record of length T resolves frequency only to $1/T$, so the coherent power a fixed-bin verifier collects is the Lorentzian mass within a half-width $B = 1/2T$ of the centre,

$$\frac{P_c(\sigma)}{P_c(0)} = \frac{2}{\pi} \arctan\left(\frac{B}{\gamma}\right) = \frac{2}{\pi} \arctan\left(\frac{1}{2\pi f_0 T \sigma^2}\right), \quad (6)$$

whose small-argument form is $1/(\pi^2 f_0 T \sigma^2)$. Matching that tail to $(1 + \kappa\sigma^2)^{-1}$ gives

$$\kappa = \pi^2 f_0 T, \quad (7)$$

a parameter-free prediction rather than a fitted scale — the width and bandwidth conventions are fixed by the explicit calculation for the rectangular-bin, untapered estimator used below, so no $O(1)$ factor is left free. The Lorentzian of Equation (3) is not a pointwise approximation to Equation (6): the two differ by up to 22% near $\sigma = 0.02$, where the arctan is still turning over, and agree to better than 1% only for $\sigma \gtrsim 0.2$. What matters is that using the Lorentzian as the fit model does not bias the constant it returns — fitting Equation (3) to Equation (6) across the σ grid swept below recovers $\kappa = 2955$, within 0.2% of $\pi^2 f_0 T$.

Proof of Proposition 3.6. Let S be the fixed-bin statistic and consider the threshold tests $\{S \geq c\}$. The advantage of the best such test is $J = \sup_c (\Pr_{\text{train}}[S \geq c] - \Pr_{\text{null}}[S \geq c])$, the *one-sided* Kolmogorov–Smirnov distance of the score laws (one-sided because training traces score high, so only that direction is a detection advantage; one-sided $\leq$ two-sided $\leq \text{TV}$, so the inequality we need still runs the right way). These are probabilities under the two laws, so J is a population quantity, and each $\{S \geq c\}$ is a deterministic single-trace test that attains its own difference — existence is all Definition 3.4 needs. Each such test is a measurable function of the trace, so by data processing $J \leq \text{TV}(P_{\text{train}}, P_{\text{null}})$. If $J(\sigma) > \varepsilon$ then total variation exceeds ε and the attack is not ε -covert by Definition 3.4; equivalently, ε -covertiness requires $J(\sigma) \leq \varepsilon$. Empirically J decreases in σ over the swept range, so the requirement becomes a lower bound $\sigma \geq \sigma^*(\varepsilon)$; we do not prove that monotonicity, which would need a stochastic-ordering argument for the score laws rather than merely the decay of the coherent power that drives them. Fitting the functional form of Equation (3) to J , $J(\sigma) = J_0/(1 + \kappa_J \sigma^2)$, and inverting gives

$$\sigma^*(\varepsilon) = \kappa_J^{-1/2} \sqrt{J_0/\varepsilon - 1}, \quad (8)$$

which grows like $\varepsilon^{-1/2}$ for small ε . That asymptotic form is a property of the *fitted* rolloff extrapolated beyond the swept range, not a derived law, which is why Proposition 3.6 claims only that σ^* is positive and increases as ε falls. Enlarging the verifier class can only increase the attainable advantage at each σ , hence only raise the class-indexed $\sigma_C^*(\varepsilon) = \inf\{\sigma : \sup_{\phi \in C} J_\phi(\sigma) \leq \varepsilon\}$; σ^* above is that quantity for the singleton fixed-bin class.

Two points of bookkeeping. First, the rolloff constant κ_J of the *advantage* is not the κ of the *coherent power* in Equation (3): the advantage saturates near one while any excess at f_0 remains separable from a line-free null, so it decays far more slowly. The two are not interchangeable in either direction — coherent power is not a detection advantage at all, so it can never be substituted into Equation (8), whatever its rolloff. Second, J is not interchangeable with the rank-based Gini index $2 \text{AUC} - 1$: the latter can exceed J , so it is not the advantage of any single-trace test and cannot appear in Equation (8). Both are reported in the summary artefact; only J carries the bound.

Discussion of Assumption 3.8, and why it is an assumption. Hold the prover’s total useful work fixed. Then a period is occupied either by training work or by idling, and period jitter has two realisations. Padding with idle time holds the real work per iteration fixed, so every period must be at least as long as the compute it contains and the mean period inflates with σ ; throughput falls, and Section 6 measures the penalty at 15–375% over the swept range. Varying the real work per iteration avoids idling and is measured at essentially zero throughput cost, but within Definition 3.3 it cannot make the schedule aperiodic: the accumulated work is an integer number of micro-steps, so the up-phase is quantised rather than continuously tunable, and the exchange down-phase keeps its duration whatever that count, leaving a component at the exchange cadence. Removing *that* requires either inserting idle time, which returns to the first realisation and its throughput cost, or desynchronising the exchange, which alters the optimisation actually performed and is therefore paid for in learning efficiency.

Two gaps stop this being a proof, and both matter.

The first is that the surviving component is not one the fixed-bin verifier can use. Section 6 measures the fixed tests under variable real work falling to detection rate 0.26 with area under the curve 0.44 — below chance — and the multitaper F-test at its floor. So the residual at the exchange cadence is recovered only by the order-tracking class, which is precisely the class Proposition 3.6 does *not* instantiate. Read strictly, then, the argument shows that a work-varying prover remains detectable *by a tracker*, not that it must pay to become ε -covert against the fixed-bin verifier whose σ^* we computed. Closing that gap needs σ^* for the tracking class, which we do not have.

The second is that fixing total useful work is a real restriction. A prover free to substitute useful *non-training* work — the additive-burial and dilution families of Section 3.2 — can modulate the period while idling neither the device nor the training job, and pays in neither training throughput nor learning efficiency. Frequency scaling is a further route the two-realisation dichotomy does not cover. So the disjunction holds over timing attacks at fixed work, not over the adversary’s full repertoire.

What the numerical check does support is the weaker comparative claim: at matched total distortion, and for every $\sigma \geq 0.05$, the work-varying attack is *more* detectable than idle-padded jitter, so substituting real-work variation does not buy coventness for free relative to the priced attack — it trades throughput cost for detectability. That comparison is measured; the cost disjunction itself is assumed, and Assumption 3.8 is labelled accordingly.

Numerical check. Figure 12 tests every load-bearing step against the generator and meter of Section 4 rather than an idealised channel. Training and inference-null populations ($n = 600$ per class, $f_0 = 1$ Hz, $T = 300$ s, $f_s = 20$ Hz, seeded) are pushed through the full observation map and scored with the fixed-bin statistic. The coherent power follows Equation (3) with $R^2 = 0.95$ and a fitted $\kappa = 3068$ against the parameter-free $\pi^2 f_0 T = 2961$ of Equation (7), with nothing fitted — but that 4% should be read as an order-of-magnitude agreement rather than a two-figure one. The linearised fit weights each point by σ^4 , so 81% of its leverage sits on the largest σ swept, and there the generator realises a cadence CV of 1.9 against a nominal 0.5: the mean period has drifted well off f_0 and the fixed bin is no longer measuring line broadening alone. Restricting the fit to the faithful regime $\sigma \leq 0.1$, where the realised CV tracks the nominal to within about 13%, gives $\kappa = 2812$, or 5% below the prediction. The honest statement is agreement at the several-percent level from either end of the sweep, which still separates $\pi^2 f_0 T$ from $\pi f_0 T$ by the factor of 3.1 that distinguishes them — the strongest single confirmation of the physical picture here.

The attained advantage requires more care, because it is estimated. Our plug-in $\hat{J}$ maximises over thresholds on the same sample it evaluates, so it is upward-biased; under H_0 at $n = 600$ its fluctuation reaches 0.093, and a measured $\hat{J}$ below that carries no evidence of separation. Because the fit linearises as $J_0/J - 1$, which diverges as $J \rightarrow 0$, such points would otherwise dominate it — at the $n = 80$ sizing we first used, two washed-out points carried 97% of the fit and κ_J varied fourfold across seeds. We therefore exclude points below the H_0 floor and report intervals. On the surviving window $\sigma \in [0.05, 0.35]$ the advantage rolls off with $\kappa_J = 32$ (95% bootstrap interval [24, 40], $R^2 = 0.90$, $J_0 = 1.0$), giving

$$\sigma^*(0.5) = 0.18 [0.16, 0.20], \quad \sigma^*(0.2) = 0.36 [0.32, 0.41],$$

and, by extrapolation beyond the swept range, $\sigma^*(0.1) = 0.53$ and $\sigma^*(0.05) = 0.78$.

Two caveats bound how far these numbers travel. First, the generator realises the nominal distortion only up to $\sigma \approx 0.2$: beyond it the reciprocal Gaussian period draw and a clamp take over, and the measured cadence coefficient of variation runs high (1.2 at $\sigma = 0.35$, 1.9 at $\sigma = 0.5$), as does D against $(2\pi)^2 \sigma^2 f_0$. Only $\sigma^*(0.5)$ falls inside that faithful region; $\sigma^*(0.2)$ and everything beyond it do not, and the artefact flags them. Second, the fit window reaches past $\sigma = 0.2$ by choice rather than necessity: the five points at $\sigma \leq 0.2$ are unsaturated and do support a fit, but the deep-rolloff point at $\sigma = 0.35$

Lower-bound spike: covertness of i.i.d. phase jitter (fixed-bin verifier, $f_0=1$ Hz, $T=300$ s)

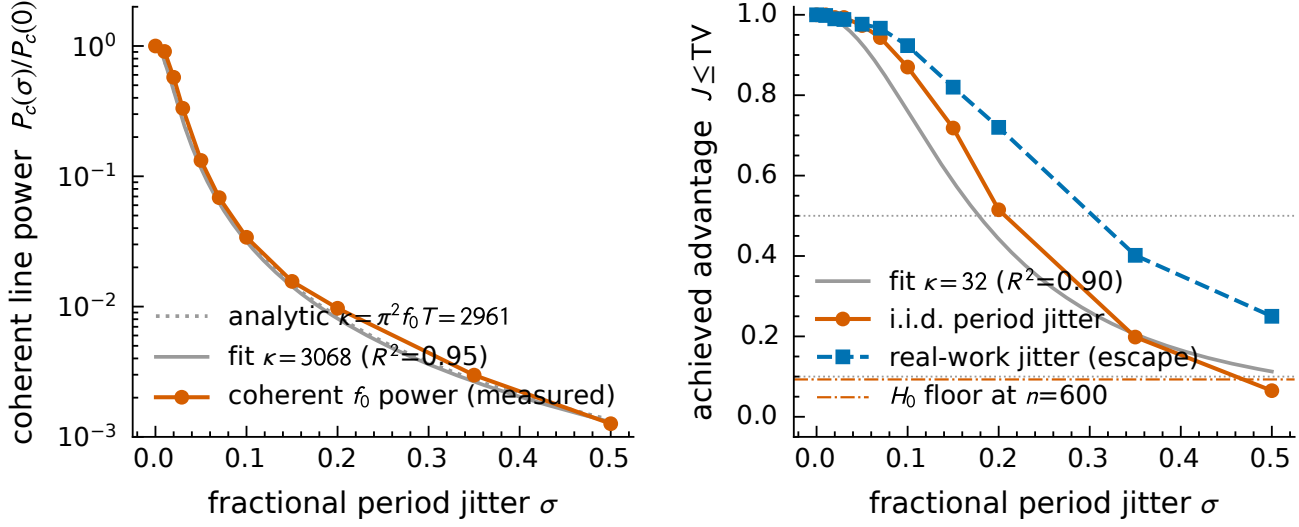


Figure 12. Numerical check on the covertness cost bound, through the full generator and meter ($n = 600$ per class). **Left:** coherent power at the nominal cadence against jitter scale. The fitted and the parameter-free analytic rollofts of Equations (3) and (7) are indistinguishable at this scale; the fitted constant agrees with the parameter-free prediction at the ten-percent level (see the numerical check). **Right:** the advantage attained by the fixed-bin threshold test, $J \leq TV$, for idle-padded period jitter and for the near-free work-varying attack; dotted lines mark $\varepsilon \in \{0.1, 0.5\}$, whose crossings give $\sigma^*(\varepsilon)$ (the $\varepsilon = 0.1$ crossing lies beyond the swept range and is reached by extrapolation). The dash-dotted line is the H_0 fluctuation of the estimator: the $\sigma = 0.5$ point falls below it and is excluded from the advantage fit shown here (it is retained in the coherent-power fit on the left, which has no sampling floor of its own). For $\sigma \geq 0.05$ the work-varying attack is the more detectable of the two at matched distortion, the comparison Assumption 3.8 appeals to; below that the two are indistinguishable within sampling error. Both arms set the period-jitter scale explicitly, so the distortion really is matched. The verifier here is fixed at the known cadence, not the order tracker of Section 5.

carries most of the leverage on κ_J . Restricting the fit to the faithful region alone gives $\kappa_J = 22$ and hence larger thresholds, $\sigma^*(0.5) = 0.22$ and $\sigma^*(0.2) = 0.43$. The thresholds are therefore good to about 20%, not to two significant figures, and the direction of that fit-window uncertainty is towards *more* required distortion, not less. Finally, the work-varying series attains a higher advantage than period jitter at every $\sigma \geq 0.05$, the two being indistinguishable within sampling error below that, which is the comparison Assumption 3.8 appeals to. Both arms set the period-jitter scale explicitly, so this is a comparison at matched total distortion.

From distortion to a price. The distortion thresholds can be read against the measured cost anchors of Section 6, which price the idle-padded realisation on the same nominal σ grid. Hiding from the fixed-bin verifier to $\varepsilon = 0.5$ needs $\sigma^* = 0.18$, which falls between the measured levels $\sigma = 0.1$ and $\sigma = 0.2$ and so between 32% and 66% throughput overhead; to $\varepsilon = 0.2$ it needs $\sigma^* = 0.36$, between 159% and 375%. Given the roughly 20% uncertainty on σ^* established above, and the scope limits of Remarks 3.9 and 3.10, the robust form of the statement is the order of magnitude rather than the bracket: driving a fixed-bin verifier to $\varepsilon = 0.5$ costs *tens of percent* of throughput, and to $\varepsilon = 0.2$ costs *of order 100% or more*. Both readings survive the fit-window sensitivity; the individual brackets do not, and the $\varepsilon = 0.5$ one in particular moves up a level under the faithful-region fit.

This is the sense in which the bound has a price, and Figure 5 is its empirical counterpart. Three things keep the two from being conflated. The thresholds come from a fixed-bin verifier at the known cadence on 600 traces per class, whereas the frontier scores a five-detector bank on 200. The pairing is by nominal σ , but the two campaigns realise the same nominal setting somewhat differently once past the faithful region — cadence coefficient of variation 0.50 here against 0.84 there at $\sigma = 0.35$ — so the mapping is exact only for $\sigma \lesssim 0.2$. And the frontier prices attacks against a detector bank that includes the tracker, whose σ^* we never computed. They are read side by side, not on shared axes.

What the bound does not cover. The propositions above concern one attack family. Making the measured frontier a statement about hiding in general would need one of two further arguments: that the swept families span what a hiding adversary can do, or a bound of the kind above for each of them. We have the second for i.i.d. period jitter and make no claim of the first, so Section 6 should be read as a sample of the attack space rather than its boundary. The choice of distributional distance is also not settled. We use total variation, reached through an attained advantage; Kullback–Leibler and Hellinger distances tensorise more conveniently over the many-iteration product law and may give a tighter closed form, which we leave open.